

θ correlations and a causal range criterion for the Weil form

Dylan Suvlu

24 September 2026

Abstract. The completed-operator construction of the Weil form yields a response kernel on positive rates, $K(r, s) = \frac{\ell(r) + \ell(s)}{r + s}$, where $\ell = \frac{M'}{M}$ and $M(r) = \frac{\xi(\frac{1}{2} + r)}{\xi(\frac{1}{2})}$. Its values are limits of localized Weil pairings on completed outputs of exponential sources. We ask whether every finite matrix of this kernel is positive semidefinite. That condition is much stronger than the positivity of ℓ' at each rate. The classical θ probability law has moment generating function M . Using it, we realize K as the correlation kernel of an explicit self-adjoint Hilbert–Schmidt Markov operator, so positivity of all finite matrices is positivity of this one operator. Through a derivative-trace criterion and a one-sided support criterion for a completed Cayley unitary, we show that this is also equivalent to membership of one explicit source b_* in the range of a compact self-adjoint Hankel operator H_b . That range is dense but not closed, so an actual preimage is required. When one exists, it is unique and causal, with squared norm $\frac{1}{2}$. Accordingly, the regularized inverse energy of b_* tends to $\frac{1}{2}$ when $b_* \in \text{Ran } H_b$ and diverges otherwise. In the second case a rank-one perturbation of H_b^2 has exactly one negative eigenvalue. Replacing the θ law by an explicit positive, even density gives positive coupling entries and a log-density concave in the squared variable, yet the matrix sign fails. By the arithmetic companion, $b_* \in \text{Ran } H_b$ is equivalent to the Riemann hypothesis, and so therefore is positivity of all finite matrices of K . Neither is decided here.

1. Introduction

The Weil explicit formula leads to a quadratic-form positivity problem in which arithmetic data and the zeros of the completed ζ function describe the same pairing [1, §V, pp. 8–9]. This paper studies the sign of a particular completed response kernel indexed by positive real rates. The question is whether every finite matrix of this kernel is positive semidefinite. We identify that sign with positivity of one compact self-adjoint operator and with solvability of one fixed Hankel equation.

Write

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s), \quad M(r) = \frac{\xi(\frac{1}{2} + r)}{\xi(\frac{1}{2})}, \quad \ell(r) = \frac{M'(r)}{M(r)}. \quad (1)$$

The classical θ probability law gives M as the moment generating function of an even positive density [2, §§2.1–2.2, (16)–(21) and Proposition 1]. For real tilts, ℓ is the mean and $\ell'(r) = \text{Var}_r(t) > 0$. This scalar inequality is much weaker than the sign problem considered here: with $K(r, s) = \frac{\ell(r) + \ell(s)}{r + s}$, we ask whether

$$\sum_{j,k} \overline{c_j} c_k K(r_j, r_k) \geq 0 \quad \text{for every finite set } r_j > 0 \text{ and } c_j \in \mathbb{C}. \quad (2)$$

The latter condition tests interactions between distinct tilts and arbitrary coefficients; it is not a consequence of positivity of the variance at each tilt.

The denominator of K has a direct source interpretation:

$$\frac{1}{r+s} = \int_0^\infty e^{-rt} e^{-st} dt. \quad (3)$$

It is the ordinary half-line Gram factor of two exponential sources. In the completed-operator construction, the numerator arises by differentiating the Gram with respect to the completion parameter. For the normalized sources used here, the derivative pairing is $K(r, s)$, which is the limit of localized Weil pairings on completed outputs as the support expands, with the sources held fixed [3, §3, Theorem “The Weil form as a leakage derivative” and Proposition “Support-compression limit”]. Section 2.1 gives the source normalization and the limiting identity. This is the completed-source meaning of the positive-rate kernel; the pairings are not being assigned directly to arbitrary unsmoothed exponentials.

Our two principal equivalences are

$$\text{all finite rate matrices are positive semidefinite} \Leftrightarrow B \geq 0 \Leftrightarrow b_* \in \text{Ran } H_b. \quad (4)$$

Here B is a compact self-adjoint Markov operator constructed explicitly from the θ law. The first equivalence replaces all finite rate tests by its spectral sign. The second replaces that sign by actual solvability for one specified source b_* under a compact self-adjoint Hankel operator H_b . Its range is dense and nonclosed, so approximation by range elements carries no such information: the criterion requires an L^2 preimage. Section 6 defines the Hankel kernel and the source explicitly.

The proof passes through three representations. First, a symmetric cross-Stein coupling has the two-rate kernel as its exponential correlation. The positive-rate exponentials are total, so testing all of them determines the sign of B . A positive coupling makes B positivity preserving and Markov, but does not make its quadratic form positive. Second, parity locking and analytic translation features express the sign as a derivative-trace inequality. The analytic totality argument is what makes local derivative data determine the global two-rate sign.

Third, graph coordinates turn the trace form into a difference of two orthogonal projections. Positivity becomes inclusion of their ranges, and Fourier transform identifies that inclusion with one-sided support invariance of a completed Cayley unitary. A fixed exponential source detects the required invariance, yielding the Hankel range condition above. This construction also identifies the regularized inverse-energy limit: it is $\frac{1}{2}$ when a preimage exists and infinity otherwise. A trace-class rank-one perturbation gives an equivalent detector with at most one negative eigenvalue.

This is the half-line branch of the completed-operator construction. The finite-characteristic companion [4] studies a different test family: periodic Fourier sections and their full least eigenstates. Its shifted finite positivity comes from subtracting the least eigenvalue, so its compactness and zero-retention theorem does not require the positive-rate condition studied here.

The probability representation is classical; the correlation operator and the passage from its sign to the distinguished source are the constructions studied here. The parity argument uses Hansen’s presentation of Löwner’s theorem [5]. The completed derivative pair and its phase already occur in Lagarias’s work [6] and in Suzuki’s Weil-distribution Hilbert spaces [7]. Our support and range statements use the full-line unitary and its dense-range construction, without assuming the innerness of that phase in a half-plane.

Sections 2–3 construct the θ law and correlation operator. Sections 4–5 prove the derivative-trace and support criteria. Section 6 proves the actual-range theorem and inverse-energy detector. Section 7 gives a Gaussian check and an exact negative example separating scalar positivity from the two-rate sign. The final section states the first-power range endpoint whose fractional regularity is studied in the arithmetic-response companion [8].

2. The critical θ law

Write $\xi(s) = s(s-1)\pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})\frac{\zeta(s)}{2}$ and $f_0 = \xi(\frac{1}{2}) > 0$. The normalized even θ density is, for $t \geq 0$,

$$p(t) = \frac{1}{f_0} \sum_{n \geq 1} (4\pi^2 n^4 e^{9\frac{t}{2}} - 6\pi n^2 e^{5\frac{t}{2}}) e^{-\pi n^2 e^{2t}}. \quad (5)$$

Poisson summation gives its smooth even continuation. Its bilateral moment generating function and logarithmic response are

$$M(r) = \int_{\mathbb{R}} e^{rt} p(t) dt = \frac{\xi(\frac{1}{2} + r)}{f_0}, \quad \ell(r) = \frac{M'(r)}{M(r)}. \quad (6)$$

In particular M is even, $M(r) > 0$ on the real line, and ℓ is odd.

The probability normalization is related to the θ laws studied by Biane, Pitman and Yor [2, §§2.1–2.2, (16)–(19), and §3.1, (27)]. Let G_n be independent Γ variables with shape 2 and rate 1, and set

$$T = \sum_{n \geq 1} \frac{G_n}{\pi n^2}. \quad (7)$$

The series converges almost surely and in L^1 . Its Laplace transform is

$$\mathbb{E} e^{-\lambda T} = \prod_{n \geq 1} \left(1 + \frac{\lambda}{\pi n^2}\right)^{-2} = \left(\frac{\sqrt{\pi\lambda}}{\sinh(\sqrt{\pi\lambda})}\right)^2. \quad (8)$$

The corresponding θ Mellin identity is $\mathbb{E} T^a = 2\xi(2a)$. Under the probability measure tilted by $T^{\frac{1}{4}}$, the variable $t = \frac{\log(T)}{2}$ has density (5). One can verify the normalization directly: the density of T is $\sum_n (4b_n^2 u - 6b_n) e^{-b_n u}$, $b_n = \pi n^2$; substitution $u = e^{2t}$ and division by $2f_0$ give (5). Integrating the θ formula in an initial right half-plane and using reciprocity extends the Mellin identity to all complex a . The exponential upper tail and reciprocal-exponential lower tail justify this continuation as an ordinary moment identity.

Consequently $\ell'(r)$ is the variance of t under its real exponential tilt. Scalar variance positivity is a different question from positivity of the two-rate kernel

$$K(r, s) = \frac{\ell(r) + \ell(s)}{r + s}, \quad r, s > 0. \quad (9)$$

The divisor $r + s$ is the Gram divisor of half-line exponentials and will be kept throughout.

We record the analytic estimates used below. The density is holomorphic in $|\Im t| < \frac{\pi}{4}$. On each closed smaller strip, p and every fixed derivative have double-exponential tails in both real directions. For the positive end, separate the first term in (5): with $\Lambda_t = \pi e^{2t}$,

$$p(t) = 4 \frac{\pi^2}{f_0} e^{9\frac{t}{2} - \Lambda_t} \left(1 - \frac{3}{2\Lambda_t} + O(e^{-3\Lambda_t})\right). \quad (10)$$

Fixed derivatives follow by differentiating the normally convergent series; reflection gives the other end. Thus, for $V = -\log p$,

$$V'(t) = 2\Lambda_t - \frac{9}{2} + O(\Lambda_t^{-1}), \quad V''(t) = 4\Lambda_t + O(\Lambda_t^{-1}). \quad (11)$$

The substitution $u = \Lambda_t(e^{2(v-t)} - 1)$ in a tail integral gives

$$\int_t^\infty p(v) dv \sim \frac{p(t)}{2\Lambda_t}, \quad \int_t^\infty vp(v) dv \sim t \frac{p(t)}{2\Lambda_t}. \quad (12)$$

These estimates also prove all the differentiation and integral domination assertions needed later. Contour shifting in the physical strip gives exponential real-frequency decay for the Fourier transforms of p , tp and their fixed derivatives.

2.1. The exponential tests and the completed Weil response

The rate kernel has a concrete source interpretation in the completed Weil construction. Let $X_+(s)$ be the backward-convolution family on $L^2(0, \infty)$ from [3, §3, “The Weil form as a leakage derivative” and “Support-compression limit”], and let a dot denote its real parameter derivative at $s = \frac{1}{2}$. Its Laplace multiplier is

$$m_s(r) = 2 \frac{\xi(s+r)}{(r+\frac{5}{2})(r+\frac{3}{2})\Gamma_{\mathbb{R}}(r+\frac{5}{2})}, \quad \Gamma_{\mathbb{R}}(z) = \pi^{-\frac{z}{2}}\Gamma\left(\frac{z}{2}\right). \quad (13)$$

For $e_r(t) = e^{-rt}$, $r > 0$, the convolution eigenvector identity gives $X_+(\frac{1}{2})e_r = m(r)e_r$ and $\dot{X}_+e_r = \ell(r)m(r)e_r$. Here $m(r) = m_{\frac{1}{2}}(r) > 0$. Put $u_r = \frac{e_r}{m(r)}$ and $\mathcal{G}_+(s) = X_+(s)^*X_+(s)$. Then

$$\langle u_r, \dot{\mathcal{G}}_+u_s \rangle = \langle \ell(r)e_r, e_s \rangle + \langle e_r, \ell(s)e_s \rangle = K(r, s). \quad (14)$$

All sources in this identity are ordinary half-line L^2 vectors; the derivative is a bounded source-coordinate operator.

The quoted support-compression result identifies the same pairing with localized Weil forms. With Q_L restriction to $(0, L)$ and $X_L = X_L(\frac{1}{2})$,

$$K(r, s) = \lim_{L \rightarrow \infty} W_L(X_L Q_L u_r, X_L Q_L u_s). \quad (15)$$

The sources are held fixed in this limit. This is the normalization behind the term Weil response: our positive-rate matrices test the derivative of the completed source Gram, which is the Weil pairing on completed outputs. It does not require a global arithmetic pairing on unsmoothed exponentials below its ordinary convergence line.

3. A cross-Stein correlation operator

We seek a symmetric coupling whose exponential moments give the two-rate kernel. The factor $r + s$ suggests differentiation in the diagonal direction: integration by parts will produce the required numerator if

$$(\partial_x + \partial_y)j(x, y) = -(x + y)p(x)p(y). \quad (16)$$

Indeed the moment of the right-hand side is $-[M'(r)M(s) + M(r)M'(s)]$, while integration by parts on the left supplies $-(r + s)$. Integrating this equation along diagonal rays, with a vanishing tail at infinity, gives the coupling below. Its marginal determines the weighted Hilbert space in which the correlation becomes an operator.

Define

$$w(x) = \int_{|x|}^\infty up(u) du, \quad \mathcal{H}_w = L^2(\mathbb{R}, w(x) dx), \quad (17)$$

$$j(x, y) = 2 \int_{\frac{|x+y|}{2}}^{\infty} v p\left(v + \frac{x-y}{2}\right) p\left(v - \frac{x-y}{2}\right) dv. \quad (18)$$

The measure $w(x) dx$ has total mass $v_0 = \int x^2 p(x) dx$, rather than one. Its normalization by v_0 is the zero-bias law of the centered variable with density p [9, Definition 1.1 and Lemma 2.1(2)].

Theorem 1 (Correlation realization).

The operator

$$(Bh)(x) = \frac{1}{w(x)} \int_{\mathbb{R}} j(x, y) h(y) dy \quad (19)$$

is a self-adjoint Hilbert–Schmidt Markov contraction on $\mathcal{H}_w$. For $\varphi_r(x) = \frac{e^{rx}}{M(r)}$,

$$K(r, s) = \langle \varphi_r, B\varphi_s \rangle_w. \quad (20)$$

The positive-rate functions φ_r , $r > 0$, are total. Hence all finite matrices $(K(r_j, r_k))$ are positive semidefinite if and only if $B \geq 0$.

Proof. Marginals and the Markov property. Put $t = \frac{x+y}{2}$, $d = \frac{x-y}{2}$. The product $p(v+d)p(v-d)$ is even in v , so the lower endpoint in (18) can equally be written t . Since $\partial_x + \partial_y = \partial_t$,

$$(\partial_x + \partial_y)j(x, y) = -(x+y)p(x)p(y). \quad (21)$$

The kernel is positive and symmetric, and $j(-x, -y) = j(x, y)$. Tonelli, the Jacobian $dx dy = 2 dt dd$, and the variables $v+d, v-d$ give

$$\int \int j = 8 \int_{\mathbb{R}} \int_0^{\infty} v^2 p(v+d)p(v-d) dv dd = v_0. \quad (22)$$

Integrating (21) in y shows that the first marginal has derivative $-xp(x)$. Integrability fixes its constant, yielding $\int j(x, y) dy = w(x)$. Symmetry gives the other marginal.

The marginal identity proves $B1 = 1$. Jensen’s inequality for $j(x, y) d\frac{y}{w(x)}$ proves $\|Bh\|_w \leq \|h\|_w$, and symmetry gives self-adjointness.

The correlation identity. Integration by parts in (21) against e^{rx+sy} gives

$$(r+s) \int \int j(x, y) e^{rx+sy} dx dy = M'(r)M(s) + M(r)M'(s). \quad (23)$$

The marginal exponential moments and Cauchy–Schwarz under $\frac{j}{v_0}$ justify every real tilt. Division gives (20).

Totality. If h is orthogonal to every positive-rate exponential, the transform $\int \overline{h(x)} w(x) e^{zx} dx$ is entire by Cauchy–Schwarz and the exponential moments of w . It vanishes on the positive real axis, hence everywhere. Fourier uniqueness gives $h = 0$. The same argument proves polynomial density by vanishing of derivatives at zero, and proves the corresponding even and odd density statements.

Hilbert–Schmidt compactness. Let $V = -\log p$ and $W_d(v) = V(v+d) + V(v-d)$. Then

$$j(x, y) = 2p(x)p(y)e^{W_d(t)} \int_{|t|}^{\infty} v e^{-W_d(v)} dv. \quad (24)$$

The function V'' is bounded below and tends to infinity in both tails, by (10). For all sufficiently large $|d|$, $W_d'' \geq 2$ everywhere. Since W_d is even, $W_d'(v) \geq 2v$ for $v \geq 0$, and the displayed ratio is at most $\frac{1}{2}$. For bounded d , the same bound holds for all sufficiently large $|t|$, using $V'(u) \geq u$ in the positive tail. On the remaining compact rectangle the ratio is bounded by continuity and uniform tail convergence. Thus $j(x, y) \leq Cp(x)p(y)$.

Finally, (12) gives $\int \frac{p(x)^2}{w(x)} dx < \infty$: its positive-tail integrand is asymptotic to $2\Lambda_x \frac{p(x)}{x}$. The squared Hilbert–Schmidt norm is $\int \int \frac{j(x, y)^2}{w(x)w(y)} dx dy$, which is finite. Boundedness and totality now give the last equivalence. $\square$

The coupling replaces an all-rank family of matrices by one compact self-adjoint operator. Its Markov property controls averages and its norm; the sign of every quadratic form remains a spectral question. For example, the symmetric Markov matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (25)$$

has eigenvalues 1 and -1 . This two-state example illustrates the distinction, rather than modeling the θ kernel. Here the identity

$$\langle h, (I - B)h \rangle_w = \frac{1}{2} \int \int j(x, y) |h(x) - h(y)|^2 dx dy \quad (26)$$

only places the spectrum of B in $[-1, 1]$. It also gives a positive-minus-positive representation of the rate kernel. If

$$P(r, s) = \frac{M'(r + s)}{(r + s)M(r)M(s)}, \quad (27)$$

then $P - K$ is the Gram of $\frac{\varphi_r(x) - \varphi_r(y)}{\sqrt{2}}$ in $L^2(j(x, y) dx dy)$. The target sign is the domination $P - K \leq P$.

4. Odd reflection and derivative traces

Reflection commutes with B , giving even and odd sectors B_+, B_- . The positive-rate parity features are $c_r = \frac{\cosh(rx)}{M(r)}$ and $o_r = \frac{\sinh(rx)}{M(r)}$. Extending the transform identity to negative rates and projecting parity gives

$$\langle c_r, Bc_s \rangle_w = \frac{r\ell(r) - s\ell(s)}{r^2 - s^2}, \quad (28)$$

$$\langle o_r, Bo_s \rangle_w = \frac{r\ell(s) - s\ell(r)}{r^2 - s^2}. \quad (29)$$

All diagonal values are defined by continuity.

The parity split is useful because these two kernels become divided differences of related scalar functions. Positivity of every divided-difference matrix is the Löwner criterion for operator monotonicity: $g(A) \leq g(B)$ whenever $A \leq B$ are positive definite matrices of any common size. The integral representation for such a function will turn the other parity kernel into a positive Gram kernel. Thus the sign passes between the two sectors through a scalar representation, not through reflection symmetry alone.

Lemma 2 (Parity locking).

$$B \geq 0 \Leftrightarrow B_+ \geq 0 \Leftrightarrow B_- \geq 0. \quad (30)$$

Proof. Let $F(x) = M(\sqrt{x})$, interpreted by the even entire series, and $R(x) = \frac{F'(x)}{F(x)}$, $S_0(x) = xR(x)$. For $x = r^2, y = s^2$, the kernels (28) and (29) are respectively

$$2\frac{S_0(x) - S_0(y)}{x - y}, \quad -2r\frac{s(R(x) - R(y))}{x - y}. \quad (31)$$

Here R, S_0 are positive real analytic on $(0, \infty)$, $R(0) = \frac{v_0}{2}$ and $S_0(0) = 0$.

The all-rank Löwner theorem and the positive operator-monotone representation [5, Theorem 3.2 and Corollary 5.1] state that a positive operator-monotone function has the form

$$g(x) = a + bx + \int_{(0, \infty)} \frac{x}{x+t} d\sigma(t), \quad a, b \geq 0, \quad (32)$$

with $\int (1+t)^{-1} d\sigma(t) < \infty$. If $B_+ \geq 0$, sector totality and the first divided-difference kernel make S_0 operator monotone. Its constant a is zero. Dividing by x gives

$$R(x) = b + \int_{(0, \infty)} \frac{1}{x+t} d\sigma(t). \quad (33)$$

The odd-sector kernel is then

$$2rs \int_{(0, \infty)} \frac{1}{(r^2+t)(s^2+t)} d\sigma(t), \quad (34)$$

a Gram kernel of the functions $\sqrt{2}\frac{r}{r^2+t}$. This is the elementary divided-difference calculation that transfers the sign.

If $B_- \geq 0$, the second kernel makes $-R$ operator monotone. Inversion reverses matrix order, so the positive function $g(x) = R(\frac{1}{x})$ is operator monotone. Its representation gives a Stieltjes representation for R , with a possible additional $\frac{b_0}{x}$ term. The finite value $R(0)$ excludes that term. Formula (33) follows again, and the even kernel is the positive Gram

$$2b + 2 \int_{(0, \infty)} \frac{t}{(r^2+t)(s^2+t)} d\sigma(t). \quad (35)$$

This proves the equivalences. The measure is constructed only under a positivity premise. $\square$

On the positive half-line let $D_o(x, y) = j(x, y) - j(x, -y)$. Positivity of B_- is equivalent to positivity of this continuous kernel: restriction and odd extension give one direction, while compactly supported approximations to point masses give the other. Define in $L^2(0, \infty) \oplus L^2(0, \infty)$

$$a_x(u) = \frac{p(u+x) - p(u-x)}{2}, \quad (36)$$

$$b_x(u) = \frac{(u+x)p(u+x) - (u-x)p(u-x)}{2}, \quad V_x = (a_x, b_x), \quad J(a, b) = (b, a). \quad (37)$$

The common-translation formula

$$j(x, y) = \int_0^\infty (x+y+2u)p(u+x)p(u+y) du \quad (38)$$

is valid even when its initial integrand is signed. Odd projection gives

$$D_o(x, y) = 2\langle V_x, JV_y \rangle. \quad (39)$$

The reflected kernel now has a bounded indefinite feature pairing: its sign is tested by J on the span of the vectors V_x . Analyticity lets us replace that whole family by derivatives at one point. The following observation specifies the topology in which this replacement is valid.

Lemma (Analytic feature totality). Let $\mathcal{E}$ be a complex Hilbert space and let $F : \Omega \rightarrow \mathcal{E}$ be holomorphic in norm on a connected open strip Ω containing the real axis. Then

$$\overline{\text{span}\{F(x) : x > 0\}} = \overline{\text{span}\{F^{(k)}(0) : k \geq 0\}}. \quad (40)$$

If $\mathcal{E}$ has a fixed conjugation and $F(x)$ belongs to its fixed real subspace for real x , the corresponding real closed spans also agree. If F is odd, only odd jets are needed.

Proof. Continuity puts $F(0)$ in the closure of the positive-real translates. The forward finite difference

$$h^{-k} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} F(jh) \quad (41)$$

converges in Hilbert norm to $F^{(k)}(0)$ as h decreases to zero. Thus every jet belongs to the first span. Conversely, a vector orthogonal to all jets gives a scalar holomorphic function $z \mapsto \langle v, F(z) \rangle$ whose Taylor coefficients at zero vanish. It vanishes near zero and then throughout the connected strip by the identity theorem. Hence the vector is orthogonal to every real translate. Equality of orthogonal complements proves (40). The same argument in the complexification proves the real statement; oddness makes the even jets zero. No convergence of a single Taylor series over the entire strip is required. $\square$

Theorem 3 (Derivative-trace criterion).

The positive-rate matrices of (9) are all positive semidefinite if and only if, for every real odd polynomial P ,

$$\mathcal{Q}(P) := \int_0^\infty [P(\partial)p(u)][P(\partial)(up(u))] du \geq 0. \quad (42)$$

Proof. The θ -strip bounds make $x \mapsto V_x$ holomorphic in the norm of $L^2(0, \infty) \oplus L^2(0, \infty)$ on $|\Im x| < \frac{\pi}{4}$: on every compact substrip they dominate each derivative in that Hilbert space. Apply (40) to this odd feature. Since J is bounded, its quadratic form is continuous on the common closed span. Positivity on all finite combinations of translates is therefore equivalent to positivity on all finite combinations of odd jets.

The identity $[P(\partial), u] = P'(\partial)$ gives the jet

$$P(\partial_x)V_x|_{x=0} = (P(\partial)p, P(\partial)(up)). \quad (43)$$

Its quadratic value in (39) is $4\mathcal{Q}(P)$. Real linear combinations of odd jets are exactly real odd polynomials. Kernel positivity, parity locking and the correlation theorem prove the assertion. $\square$

Writing $F_P = P(\partial)p$ and $G_P = P'(\partial)p$, the form is

$$\mathcal{Q}(P) = \int_0^\infty (uF_P^2 + F_PG_P) du. \quad (44)$$

The endpoint term cannot be discarded just because $F_P(0) = 0$. For example $\mathcal{Q}(z) = \int_0^\infty u(p')^2 du - \frac{p(0)^2}{2}$. Higher derivatives give a corresponding bulk-minus-boundary form; the analytic feature argument is what makes the entire family of such traces a global criterion.

5. Equal Grams and causal scattering

The derivative trace is a half-line pairing of a function and its logarithmic-derivative response. We will put that pairing into graph coordinates where it is the quadratic form of a difference of two orthogonal projections. For projections P, Q ,

$$P - Q \geq 0 \iff \text{Ran } Q \subset \text{Ran } P. \quad (45)$$

Necessity follows by testing on $\ker P$, and sufficiency follows from the orthogonal decomposition of nested ranges. This converts the sign question into subspace inclusion; Fourier transform will identify that inclusion with preservation of a physical half-line.

The three Fourier conventions used in this passage are:

Transform	Definition and role
$\mathcal{F}_- f(x)$	$(2\pi)^{-\frac{1}{2}} \int f(t) e^{-ixt} dt$; Hilbert-space coordinates
$\chi(x)$	$\int p(t) e^{-ixt} dt$; scalar characteristic
$\mathcal{F}_+ f(x)$	$\int f(t) e^{ixt} dt$; convolution formulas in Section 6

The negative physical half-line corresponds to the upper Hardy space under $\mathcal{F}_-$. Reflection at the end of this section produces the positive-time solution used in the range criterion.

With these conventions, on the real line χ, χ' are real, and χ is even. They have exponential decay by the θ strip. The phase S defined below extends across isolated common zeros by cancelling the common Taylor factor in its numerator and denominator. The logarithmic-derivative multiplier $\frac{\chi'}{\chi}$ retains its maximal operator domain; it is not being declared bounded at zeros of χ .

The maximal multiplier

$$T = \mathcal{F}_-^{-1} M_{i\frac{\chi'}{\chi}} \mathcal{F}_-, \quad \text{Dom } T = \left\{ f \in L^2 : \left(\frac{\chi'}{\chi} \right) \mathcal{F}_- f \in L^2 \right\} \quad (46)$$

is densely defined, closed and skew-adjoint. For a polynomial P ,

$$T[P(\partial)p] = P(\partial)(tp). \quad (47)$$

Put $v = \chi^2 + \chi'^2$. The graph norm on this orbit is $(2\pi)^{-1} \int |P(ix)|^2 v(x) dx$. Polynomials are dense in $L^2(v dx)$: if a vector is orthogonal to them, its weighted bilateral Laplace transform is analytic in a strip because v has an exponential moment; all derivatives at zero vanish, so Fourier uniqueness applies. The same proof works in each parity sector. It proves the graph-core assertion rather than merely density in an unweighted space.

Let $\mathcal{H}$ denote the Hilbert transform in the frequency coordinate, with kernel $\frac{1}{[\pi(x-y)]}$, and $\Pi = \frac{I+i\mathcal{H}}{2}$. Define

$$t_0 = \frac{\chi}{\sqrt{v}}, \quad r_0 = \frac{\chi'}{\sqrt{v}}, \quad \omega = t_0 + ir_0, \quad S = \frac{\chi - i\chi'}{\chi + i\chi'}. \quad (48)$$

The phase is unimodular on the real line. For a real polynomial P , put

$$g_P = P(ix) \frac{\sqrt{v(x)}}{\sqrt{2\pi}}, \quad h_P = \bar{\omega} g_P. \quad (49)$$

The derivative trace satisfies

$$\Re \mathcal{Q}(P) = \frac{1}{4} \langle h_P, [\Pi - M_S \Pi M_{\bar{S}}] h_P \rangle. \quad (50)$$

For odd P this is (42). To check the constants, Fourier conjugation of multiplication by $\text{sgn}(t)$ is $-i\mathcal{H}$. The transforms in (47) are $t_0 g_P$ and $ir_0 g_P$. The real full-line pairing vanishes, so its signed pairing is twice its half-line pairing. It follows that $\Re \mathcal{Q}(P) = \frac{\langle g_P, C_0 g_P \rangle}{2}$, where

$$C_0 = \frac{M_{t_0} \mathcal{H} M_{r_0} - M_{r_0} \mathcal{H} M_{t_0}}{2}. \quad (51)$$

Expansion gives

$$M_\omega \Pi M_{\bar{\omega}} - M_{\bar{\omega}} \Pi M_\omega = 2C_0. \quad (52)$$

Conjugation proves (50).

Theorem 4 (Support criterion).

Let $\mathcal{U} = \mathcal{F}_-^{-1} M_S \mathcal{F}_-$, and let P_-, P_+ be physical projection onto the negative and positive half-lines. The following are equivalent:

- all positive-rate matrices (9) are positive semidefinite;
- $P_- - \mathcal{U} P_- \mathcal{U}^* \geq 0$;
- $P_+ \mathcal{U} P_- = 0$;
- $P_+ \mathcal{U} e = 0$, where $e(t) = e^t \mathbf{1}_{t < 0}$.

Proof. The upper-half-plane Hardy projection [10, Theorems 1.1–1.3] obeys $\mathcal{F}_-^{-1} \Pi \mathcal{F}_- = P_-$. Positivity of the projection difference implies the odd trace criterion by (50). For the converse, odd trace positivity first gives $B_- \geq 0$, and parity locking gives $B \geq 0$. By approximating point masses with compactly supported tests, the continuous kernel $j(x, y)$ is positive semidefinite. Use now the unprojected feature

$$\Phi_x(u) = (p(u+x), (u+x)p(u+x)), \quad u > 0. \quad (53)$$

Its common-translation identity is $j(x, y) = \langle \Phi_x, J \Phi_y \rangle$. Thus J is nonnegative on the real span of these features. Applying (40) to Φ , rather than to the odd feature V , gives positivity on every real polynomial jet

$$P(\partial_x) \Phi_x|_{x=0} = (P(\partial)p, P(\partial)(up)). \quad (54)$$

The J -quadratic value is $2\mathcal{Q}(P)$ for this unprojected feature. Hence the trace is nonnegative for all real polynomials, not only odd ones.

To pass to the full graph space, consider the real involution $g(x) \mapsto \overline{g(-x)}$. The vectors $P(ix)\sqrt{v(x)}$ for real polynomials are dense in its fixed subspace of $L^2(dx)$: this is the real version of the weighted polynomial-density argument above, after multiplication by $\sqrt{v}$. Multiplication by $\bar{\omega}$ preserves this subspace because $\omega(-x) = \overline{\omega(x)}$. The projection difference in (50) also preserves it, as seen in physical coordinates from $S(-x) = \overline{S(x)}$ and the reality of both support projections. Continuity therefore extends positivity from the polynomial vectors h_P to this whole real space. Complexification gives positivity on the entire auxiliary Hilbert space: for a self-adjoint operator preserving that real space, the mixed imaginary terms in its quadratic form cancel. This proves the first equivalence.

Apply (45) with $P = P_-$ and $Q = \mathcal{U} P_- \mathcal{U}^*$. The range inclusion says exactly that $\mathcal{U}$ preserves the negative half-line. In particular, if $L_0 = P_+ \mathcal{U} P_-$, then

$$P_+(P_- - \mathcal{U} P_- \mathcal{U}^*) P_+ = -L_0 L_0^*. \quad (55)$$

Thus any nonzero wrong-way block produces a negative direction.

The backward translates $e(t+a)$, $a \geq 0$, are total in $L^2(-\infty, 0)$. After reflection, orthogonality would say $\int_a^\infty \overline{f(t)} e^{-(t-a)} dt = 0$ for all $a \geq 0$. This locally absolutely continuous function has derivative equal to itself minus $\overline{f(a)}$, so $f = 0$. Since $\mathcal{U}$ commutes with translations, support invariance on this one source implies it on the dense translation span and then on the whole half-line. $\square$

The graph calculation has identified a boundary phase. The completed operator pair supplies the same unitary directly, with its domain cost visible: equal Grams give an isometry between dense ranges, so no bounded inverse is required. Specifically, use the completed pair [3, §3, Corollary “Completed-output projection defect”]. With $\Gamma_{\mathbb{R}}(z) = \pi^{-\frac{z}{2}} \Gamma(\frac{z}{2})$, define the full-line family by the multipliers, at $w = -ix$,

$$m_s(w) = 2 \frac{\xi(s+w)}{\left(\frac{5}{2}+w\right)\left(\frac{3}{2}+w\right)\Gamma_{\mathbb{R}}\left(\frac{5}{2}+w\right)}. \quad (56)$$

The family is bounded and holomorphic in operator norm on a complex neighborhood of $s = \frac{1}{2}$; Appendix A gives the Euler/Stirling bounds and parameter differentiation that justify this assertion. Let X be the critical operator and $\dot{X}$ its parameter derivative. Their multipliers are $d(x)\chi(x)$ and $id(x)\chi'(x)$, with

$$d(x) = 2 \frac{f_0}{\left(\frac{5}{2}-ix\right)\left(\frac{3}{2}-ix\right)\Gamma_{\mathbb{R}}\left(\frac{5}{2}-ix\right)}. \quad (57)$$

Since χ and χ' are real on the real line,

$$X^* \dot{X} + \dot{X}^* X = 0, \quad A_+^* A_+ = A_-^* A_- = X^* X + \dot{X}^* \dot{X}, \quad A_{\pm} = X \pm \dot{X}. \quad (58)$$

The multipliers $d(\chi \pm i\chi')$ are nonzero almost everywhere. Thus A_+, A_- are injective with dense full-line ranges, and $A_+ f \mapsto A_- f$ extends from their ranges to a unitary. Its multiplier is exactly (48). This constructs $\mathcal{U}$ without a bounded inverse of A_+ .

Our support convention can be summarized as follows: the negative physical half-line corresponds to the upper Hardy space under $\mathcal{F}_-$; the source e has transform $\frac{i}{\sqrt{2\pi}(x+i)}$. The positive-time convolution solution in the next section is the reflected function $y(t) = (\mathcal{U}e)(-t)$.

6. One source and an actual Hankel range

On the full line define $a(t) = (1-t)p(t)$ and $b(t) = (1+t)p(t) = a(-t)$, and let $\kappa(t) = e^{-t}\mathbf{1}_{t>0}$. The plus-sign unnormalized Fourier transforms are

$$\hat{a}_+(x) = \chi(x) + i\chi'(x), \quad \hat{b}_+(x) = \chi(x) - i\chi'(x), \quad \hat{\kappa}_+(x) = \frac{i}{x+i}. \quad (59)$$

Thus the full-line convolution equation

$$a * y = \kappa * b \quad (60)$$

has the unique L^2 solution

$$\hat{y}_+(x) = i \frac{S(x)}{x+i}, \quad \|y\|_2^2 = \frac{1}{2}. \quad (61)$$

Existence follows from unimodularity of S and Plancherel; uniqueness follows because $\hat{a}_+$ is a nonzero entire function and hence nonzero almost everywhere. All convolutions are defined by L^1 kernels on L^2 . Equation (61) also verifies $y(t) = (\mathcal{U}e)(-t)$.

On ordinary $\mathcal{H}_+ = L^2(0, \infty)$ define

$$(H_a f)(t) = \int_0^\infty a(t+u)f(u) du, \quad (H_b f)(t) = \int_0^\infty b(t+u)f(u) du, \quad (62)$$

and the distinguished source

$$b_* = H_a \kappa, \quad b_*(t) = e^t \int_t^\infty (1-u)e^{-u} p(u) du. \quad (63)$$

The operators are self-adjoint and Hilbert–Schmidt, since, for example,

$$\|H_b\|_{\text{HS}}^2 = \int_0^\infty u(1+u)^2 p(u)^2 du < \infty. \quad (64)$$

Both are injective. If $H_b f = 0$, then $F(z) = \int_0^\infty b(z+u)f(u) du$ is holomorphic in the connected θ strip by uniform L^2 tail domination. It vanishes on the positive real axis, hence everywhere in the strip. On the real line it is convolution of b with the reflected zero extension of f . Fourier transformation and nonvanishing almost everywhere of $\chi + i\chi'$ force $f = 0$. The same proof applies to H_a . Self-adjointness gives dense range; compactness on this infinite-dimensional space makes that dense range nonclosed.

Theorem 5 (Range and causality).

The sign conditions of Section 5 are equivalent to

$$b_* \in \text{Ran } H_b. \quad (65)$$

If $H_b q = b_*$ with $q \in \mathcal{H}_+$, then the preimage is unique and $\|q\|^2 = \frac{1}{2}$.

Proof. For $t > 0$, set $y_+(t) = y(t)$ and $y_-(t) = y(-t)$. Splitting (60) at zero gives

$$W_a y_+ + H_a y_- = W_b \kappa, \quad H_b y_+ + W_b y_- = H_a \kappa, \quad (66)$$

where $W_a f(t) = \int_0^\infty a(t-u)f(u) du$ and $W_b = W_a^*$. If y is causal, the second equality gives (65).

Conversely, suppose $H_b q = b_*$. Extend q by zero to negative time. The residual $a * q - \kappa * b$ is holomorphic in the θ strip by the same uniform tail argument. At $t = -r < 0$ it is $(H_b q - H_a \kappa)(r) = 0$. Analytic continuation makes the residual vanish on the entire real line. Full-line uniqueness identifies this extension with y , proving causality and the norm assertion in (61). The support criterion completes the equivalence. $\square$

Range closure would carry no information here: it is already the whole half-line space. Nor can one simply invert W_a . A fixed smooth compact half-line vector modulated to high frequency has full-line convolution by a tending to zero in norm, by decay of its Fourier multiplier; restriction cannot increase the norm. Thus W_a is not bounded below.

Put $A = H_b^2$ and, for $\lambda > 0$,

$$\mathcal{E}(\lambda) = \langle b_*, (A + \lambda I)^{-1} b_* \rangle. \quad (67)$$

The spectral theorem makes this a positive decreasing Stieltjes function. Its measure belongs to this particular squared Hankel operator; it is not the logarithmic response of (6).

If a Hankel preimage exists, the support theorem identifies it with the causal full-line solution. That solution is the reflected unitary image of the prescribed exponential, whose squared norm is $\frac{1}{2}$. Thus any finite limit of the regularized inverse energy must be exactly $\frac{1}{2}$: the source normalization fixes its value, while failure of actual range membership forces divergence.

Theorem 6 (Inverse energy and rank-one detection).

The limit of $\mathcal{E}(\lambda)$ as λ decreases to zero is $\frac{1}{2}$ if (65) holds, and $+\infty$ otherwise. Moreover

$$\mathcal{C} = \frac{1}{2}H_b^2 - |b_*\rangle\langle b_*| \quad (68)$$

is positive semidefinite precisely in the first case. Otherwise it has exactly one negative eigenvalue, simple, equal to $-\frac{\lambda_*}{2}$, where $\mathcal{E}(\lambda_*) = \frac{1}{2}$.

Proof. If $b_* = H_b q$, functional calculus gives $\mathcal{E}(\lambda) = \langle q, A(A + \lambda)^{-1} q \rangle \rightarrow \|q\|^2 = \frac{1}{2}$. Conversely, if these energies remain bounded, let $q_\lambda = H_b(A + \lambda)^{-1} b_*$. Then

$$\|q_\lambda\|^2 = \langle b_*, A(A + \lambda)^{-2} b_* \rangle \leq \mathcal{E}(\lambda). \quad (69)$$

A weakly convergent subsequence has limit q , while $H_b q_\lambda = A(A + \lambda)^{-1} b_* \rightarrow b_*$ strongly, since A is injective. Thus $H_b q = b_*$.

If $b_* = H_b q$ with $\|q\|^2 = \frac{1}{2}$, Cauchy–Schwarz gives $|\langle b_*, f \rangle|^2 \leq \frac{\|H_b f\|^2}{2}$, so $\mathcal{C} \geq 0$. Conversely this inequality defines a bounded functional on the dense range of H_b ; Riesz representation gives a preimage of b_* . A positive operator minus a rank-one operator has negative index at most one, because any two-dimensional negative subspace would meet $b_*^\perp$. In the failure case there is a negative direction; compact self-adjoint spectral theory therefore gives one simple negative eigenvalue. Finally, $\mathcal{C}f = -\nu f$ is equivalent to $(A + 2\nu)f = 2b_*\langle b_*, f \rangle$, so a nonzero solution exists exactly when $2\mathcal{E}(2\nu) = 1$. Strict decrease from infinity to zero proves the asserted unique value. $\square$

7. Two examples separating positivity notions

For the centered Gaussian density of variance σ^2 , $tp = -\sigma^2 p'$. For odd P , if $F = P(\partial)p$, then $P(\partial)(tp) = -\sigma^2 F'$, whence

$$\mathcal{Q}(P) = -\frac{\sigma^2}{2} [F(t)^2]_0^\infty = 0. \quad (70)$$

The corresponding phase is $S(z) = \frac{1+i\sigma^2 z}{1-i\sigma^2 z}$ and has the causal orientation. At variance one the full-line solution is

$$y(t) = (2t - 1)e^{-t} \mathbf{1}_{t>0}. \quad (71)$$

It is causal and signed, with squared norm $\frac{1}{2}$. The odd trace saturates; the even-sector form need not vanish.

For a separating example, let φ_G be the standard Gaussian density and

$$p_a(t) = \frac{(t^4 + 20t^2 + 200)\varphi_G(t)}{223}. \quad (72)$$

This density is positive, even, normalized and analytic, with Gaussian tails, sufficient for the translation-feature and derivative-trace calculations here. Its squared-coordinate logarithm is strictly concave:

$$\frac{d^2}{du^2} \log p_a(\sqrt{u}) = -2 \frac{u(u + 20)}{(u^2 + 20u + 200)^2} < 0 \quad (u > 0). \quad (73)$$

Nevertheless the derivative trace is negative for $P(z) = z + z^3$:

$$\mathcal{Q}_{p_a}(P) = -\frac{20}{\pi 223^2}. \quad (74)$$

Appendix A computes this value by Gaussian moments. The cross-Stein kernel still has positive entries and a positive Markov interpretation. These facts and scalar concavity therefore do not supply the all-rank sign.

8. The critical range question

The constructions above reduce the literal sign problem to the actual range condition $H_a \kappa \in \text{Ran } H_b$. Every object in this statement is independently specified by the θ density and the fixed causal exponential. A finite bound on its regularized inverse energy would suffice; an arbitrary accurate approximation by range elements would not.

The phase S is the derivative-pair phase of completed ζ : with $E_\xi(z) = \xi(\frac{1}{2} - iz) + \xi'(\frac{1}{2} - iz)$, it is $\frac{E_\xi^\#}{E_\xi}$. Its relation to half-plane inner functions and de Branges spaces is established in the work of Lagarias [6, Theorem 1] and Suzuki [7, (1.3), Theorem 1.1, and §2.3]. The present correlation, graph and range formulas identify this endpoint through explicit physical kernels. They do not supply the literal positivity premise needed to make that phase inner.

A companion article [8] constructs the source-generated causal response of exponential order directly from integer delays, and studies finite Möbius reconstruction and the fractional ranges of $|H_b|$. That construction gives a supported response q_c before ordinary square integrability is known. Its endpoint is exactly $b_* \in \text{Ran } H_b \Leftrightarrow q_c \in L^2(0, \infty)$: then q_c is the unique preimage and has squared norm $\frac{1}{2}$. The companion's §7 identifies it with the ordinary full-line solution (61) by uniqueness on a common safe transform line. Its §2 also gives absolutely convergent Möbius and Euler kernel transports preserving this same preimage. The range and support arguments proved here do not require that additional kernel representation.

The companion's §9 shows further that this endpoint is equivalent to the Riemann hypothesis [8, §9, Corollary “The range endpoint and RH”]. By the correlation theorem of Section 3 and the range theorem of Section 6, positive semidefiniteness of every finite positive-rate matrix is therefore also equivalent to RH. This identifies the strength of the range condition; it does not decide it.

A Two supporting calculations

A.1 The completed multiplier family

These multipliers are bounded uniformly on a complex parameter neighborhood. Put $T = 2 + |x|$. Euler summation with cutoff T gives

$$\zeta(s - ix) = O(T^{1-\Re s}) \quad \left(\left| s - \frac{1}{2} \right| \leq \frac{1}{8} \right). \quad (75)$$

Stirling's formula bounds the ratio $\frac{\Gamma(\frac{s-ix}{2})}{\Gamma(\frac{\frac{5}{2}-ix}{2})}$ by $O(T^{-\frac{15}{16}})$ on this disk; the remaining polar ratio is bounded. Consequently

$$m_s(-ix) = O(T^{-\frac{5}{16}}). \quad (76)$$

These estimates follow directly from Euler summation and the uniform Γ asymptotic [11, (25.2.8) and §5.11]. Cauchy's estimates on a smaller parameter disk bound the parameter derivative and the second-order remainder uniformly in x . Thus the associated multiplier operators depend holomorphically on s in operator norm. These bounds justify the operator-norm holomorphy used in Section 5. They concern the parameter derivative of the completed multiplier, without estimating the potentially singular quotient $\frac{\zeta'}{\zeta}$ on the critical strip.

A.2 The negative derivative trace

For the density and polynomial in Section 7, with the common factor $\frac{e^{-\frac{t^2}{2}}}{223\sqrt{2\pi}}$ removed, the two differentiated factors are

$$F = -t^7 - 6t^5 - 84t^3 + 224t, \quad (77)$$

$$H = -t^8 - 3t^6 - 50t^4 + 580t^2 - 280. \quad (78)$$

Their product is

$$t^{15} + 9t^{13} + 152t^{11} - 252t^9 + 328t^7 - 58240t^5 + 153440t^3 - 62720t. \quad (79)$$

Using $\int_0^\infty t^{2k+1} e^{-t^2} dt = \frac{k!}{2}$, its integral is -40 , proving (74).

References

- [1] Bombieri, Enrico, Problems of the Millennium: the Riemann hypothesis, <https://www.claymath.org/wp-content/uploads/2022/05/riemann.pdf>. Section V, pp. 8–9 of the online PDF.
- [2] Biane, Philippe, Pitman, Jim, and Yor, Marc, Probability laws related to the Jacobi theta and Riemann zeta functions, and Brownian excursions, *Bulletin of the American Mathematical Society* **38** no. 4 (2001) 435–465, doi:10.1090/S0273-0979-01-00912-0. Preprint equation locators refer to version 1, 21 December 1999.
- [3] Suvlu, Dylan, An operator-theoretic representation of the Weil quadratic form, 2026. Companion manuscript, version v0.1.0-draft.11, revised 24 September 2026.
- [4] Suvlu, Dylan, Constant-mode transfer and compactness criteria for finite Weil characteristics, 2026. Companion manuscript, version v0.1.0-draft.4, revised 24 September 2026.
- [5] Hansen, Frank, The fast track to Löwner's theorem, *Linear Algebra and its Applications* **438** (2013) 4557–4571, doi:10.1016/j.laa.2013.01.022. Version 3, 3 February 2013.
- [6] Lagarias, Jeffrey C., Hilbert spaces of entire functions and Dirichlet L-functions, in *Frontiers in Number Theory, Physics, and Geometry I*, Springer, 2006, pp. 365–377. Theorem 1; author preprint, 2 January 2006, pp. 9–10.
- [7] Suzuki, Masatoshi, On the Hilbert space derived from the Weil distribution, *Canadian Journal of Mathematics* (2025) 1–24, doi:10.4153/S0008414X25101739. Online publication; arXiv version 3, 7 November 2025.
- [8] Suvlu, Dylan, Arithmetic reconstruction and endpoint regularity of the completed ζ response, 2026. Companion manuscript, version v0.1.0-draft.4, revised 24 September 2026.
- [9] Goldstein, Larry, and Reinert, Gesine, Stein's method and the zero bias transformation with application to simple random sampling, *The Annals of Applied Probability* **7** no. 4 (1997) 935–952. Definition 1.1 and Lemma 2.1(2); author preprint version 1, 28 October 2005.
- [10] Qian, Tao, Xu, Yuesheng, Yan, Dunyan, Yan, Lixin, and Yu, Bo, Fourier spectrum characterization of Hardy spaces and applications, *Proceedings of the American Mathematical Society* **137** no. 3 (2009) 971–980, doi:10.1090/S0002-9939-08-09544-0.
- [11] National Institute of Standards and Technology, NIST Digital Library of Mathematical Functions, <https://dlmf.nist.gov/>. Equations 25.2.8, 5.11.2 and 5.11.13.